Xinhuan SHU

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Education

Sept 2017 - Present **Ph.D. in Computer Science**
The Hong Kong University of Science and Technology, Hong Kong
- Supervisor: Prof. Huamin Qu
- Research Interest: data-driven storytelling, visual analytics

Sept 2013 - July 2017 **B.E. major in Computer Science, minor in Public Administration**
Zhejiang University, Hangzhou, China
- Supervisor: Prof. Yingcai Wu

Research / Working Experience

April 2020 - Present **Zhejiang Lab**, Hangzhou, China
Research Intern (advised by Prof. Yingcai Wu and Prof. Wei Chen), Research Center for Integrated Intelligence
- Collaborate on designing Nebula, a grammar for coordinating visualization in multiple views

Dec 2018 - Mar 2019 **Zhejiang University**, Hangzhou, China
Visiting Research Student (advised by Prof. Yingcai Wu), ZJU IDG Group
- Design DancingWords for exploring animated word clouds for data-driven storytelling

Projects

Data-Driven Storytelling

Oct 2020 - April 2021 Project Authoring tool for data-GIFs (submitted to IEEE VIS'21)
- Design DataGifify, a lightweight authoring tool for data-GIFs, inspired by short video effects and editing tool (e.g., TikTok).
- Follow a preview-based authoring paradigm that supports faceted browsing of dynamic previews with various narrative effects.

Oct 2019 - April 2020 Project Data-GIF[1] (published on IEEE VIS'20 full paper)
- Identify the emerging genre of data-driven storytelling, namely, data-GIF, and present its design space.
- Conclude with a list of design suggestions for creating more effective data-GIFs through a set of user studies.

Feb 2020 - June 2020 Project Animated Transition[2] (published on IEEE VIS'20 short paper)
- Propose a taxonomy of animated transtions in data videos in terms of how they use visual variables to build fluent narratives.

Dec 2018 - Mar 2019 Project DancingWords[3] (published on ChinaVis'20 Honorable Mention)

- Present a design space of animated word clouds for telling stories based on an exploratory workshop, including three narrative dimensions and their designs.
- Design DancingWords, a prototype tool that allows users to interactively explore narratives and generate animation with automatic layouts.

Grammar Design for Multiple Coordinated Views

April 2020 - Oct 2020 Project Nebula[4] (published on IEEE TVCG)

- Design Nebula, a natural language based grammar for coordinating visualization in multiple views based on a demonstration-driven coordination framework.

Visual Analytics for Video Data

Jan 2018 - Aug 2019 Project EmotionCues[6] (published on IEEE TVCG)

- Design EmotionCues, a visual analytics system that provides automatic multi-level visual summarization of classroom videos from the perspective of emotion.
- Integrate model uncertainty into emotion analysis and provide visual cues for reasoning of emotion evolution.

May 2019 - Aug 2019 Project AI-powered Visual Analytics Solutions for Communication and Presentation Coaching (Lab Project)

- Develop AI-based visual analytics system to help both coaches and public speaking trainees to conduct better presentation training.
- Participate as a developer to implement web-based visual interfaces for analyzing and understanding audio information of the presentations.

Interactive Visualization for Financial Data

Dec 2017 - Mar 2018 Project BitExtract[7] (published on IEEE VIS'18 full paper)

- Develop BitExtract, a visual analytics system based on the Bitcoin transaction data that summarizes the evolution of the Bitcoin market.

Sports Data Analytics

Nov 2016 - Mar 2017 Project iTTVis[9] (published on IEEE VIS'17 full paper)

- Design iTTVis, an interactive visualization system to analyze a table tennis match from time-oriented, statistical, and tactical aspects.

Publication

[1] What Makes a Data-GIF Understandable?
Xinhuan Shu, Aoyu Wu, Junxiu Tang, Benjamin Bach, Yingcai Wu, and Huamin Qu
IEEE Transactions on Visualization and Computer Graphics, InfoVis 2020

[2] Narrative Transitions in Data Videos
Junxiu Tang, Lingyun Yu, Tan Tang, Xinhuan Shu, Lu Ying, Yuhua Zhou, Peiran Ren, Yingcai Wu
IEEE VIS Conference Short Paper, 2020

[3] DancingWords: Exploring Animated Word Clouds to Tell Stories
Xinhuan Shu, Jiang Wu, Xinke Wu, Hongye Liang, Weiwei Cui, Yingcai Wu, and Huamin Qu
Journal of Visualization, ChinaVis 2020 [**Best Paper Honorable Mention**]

[4] Nebula: A Coordinating Grammar of Graphics
Ran Chen, Xinhuan Shu, Jiahui Chen, Di Weng, Junxiu Tang, Siwei Fu, Yingcai Wu
IEEE Transactions on Visualization and Computer Graphics, 2021

[5] Interactive Visual Exploration of Longitudinal Historical Career Mobility Data
Yifang Wang, Hongye Liang, Xinhuan Shu, Jiachen Wang, Ke Xu, Zikun Deng, Cameron Dougall Campbell, Bijia Chen, Yingcai Wu, and Huamin Qu
IEEE Transactions on Visualization and Computer Graphics, 2021

[6] EmotionCues: Emotion-Oriented Visual Summarization of Classroom Videos
Haipeng Zeng, Xinhuan Shu, Yanbang Wang, Yong Wang, Liguo Zhang, Ting-Chuen Pong, and Huamin Qu
IEEE Transactions on Visualization and Computer Graphics, 2020

[7] BitExTract: Interactive Visualization for Extracting Bitcoin Exchange Intelligence
Xuanwu Yue, Xinhuan Shu, Xinyu Zhu, Xinnan Du, Zheqing Yu, Dimitrios Papadopoulos, and Siyuan Liu
IEEE Transactions on Visualization and Computer Graphics, VAST 2018

[8] Pulse: Toward a Smart Campus by Communicating Real-time Wi-Fi Access Data
Aoyu Wu, Bon Kyung Ku, Furui Cheng, Xinhuan Shu, Abishek Puri, Yifang Wang, and Huamin Qu
Workshop on Visualization for Communication, IEEE VIS Conference 2018

[9] iTTVis: Interactive Visualization of Table Tennis Data
Yingcai Wu, Ji Lan, Xinhuan Shu, Chenyang Ji, Kejian Zhao, Jiachen Wang, and Hui Zhang.
IEEE Transactions on Visualization and Computer Graphics, InfoVis 2017

Awards

2017-2020	HKUST Scholarships
2017	Outstanding Graduates of Zhejiang University
2016	Honorable Mention Award for 2016 MCM/ICM
2015	University Scholarships, Zhejiang University Honorable Title of Excellent Student, Zhejiang University
2014	University Scholarships, Zhejiang University Honorable Title of Excellent Student, Zhejiang University

Invited Talks

Oct 2020	DancingWords: Exploring Animated Word Clouds to Tell Stories ChinaVis, Xi'an, China
Oct 2020	What Makes a Data-GIF Understandable? IEEE VIS Conference, Salt Lake City, USA

Services

Reviewer of	ACM Conference on Human Factors in Computing Systems (CHI), 2021 IEEE Pacific Visualization (PacificVis) Symposium, 2021

EuroVis 2021
IEEE VIS Conference, 2021

Volunteer IEEE VIS Conference, Salt Lake City, USA, 2020

Teaching *Teaching Assistant Coordinator* at Department of Computer Science and Technology, HKUST, 2019
Teaching Assistant, Introduction to Computer Science, HKUST

Skills

Programming Javascript, Python, HTML, C/C++

Web Devs Vue, React, D3, Flask, MongoDB

Languages Chinese Mandarin (native), English (professional working proficiency)

Last updated on May 6, 2021